FORM TP 2021025

TEST CODE 01238020

JANUARY 2021

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE® EXAMINATION

PHYSICS

Paper 02 – General Proficiency

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in TWO sections. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. Where appropriate, ALL WORKING MUST BE SHOWN in this booklet.
5. You may use a silent, non-programmable calculator to answer questions, but you should note that the use of an inappropriate number of significant figures in answers will be penalized.
6. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
7. **If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Copyright © 2020 Caribbean Examinations Council
All rights reserved.

Scanned with CamScanner

SECTION A

Answer ALL questions.

1. Figure 1 shows the apparatus which was used by two students to carry out an experiment to determine the focal length, f, of a lens by using the object distance, u, and the image distance, v.

At the end of the experiment the object distance, u, and the image distance, v, were recorded in Table 1.

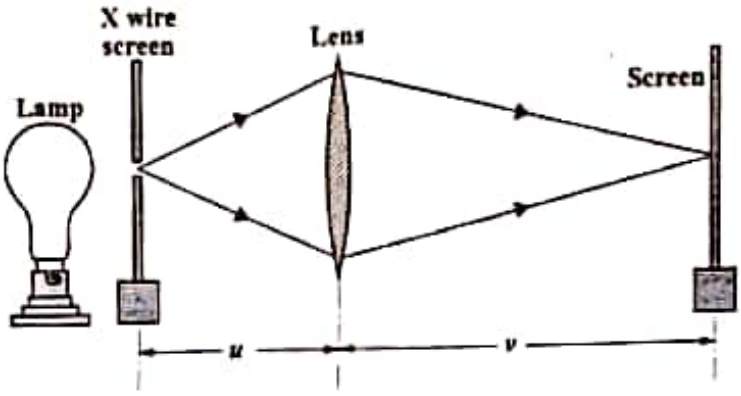


Figure 1. Experiment to measure the focal length, f, of a lens

TABLE 1: RESULTS OF THE EXPERIMENT

u (cm)	$1/u$ (cm^{-1}) × 10^{-2}	v (cm)	$1/v$ (cm^{-1}) × 10^{-2}
20.0	5·0	60.0	1·6
30.0	3.3	30.0	3.3
35.0	2·8	26.3	3.8
40.0	2.5	24.0	4.2
45.0	2.2	22.5	4·4
55.0	1·8	20.6	4.9

(a) Complete Table 1 by calculating the missing values. **(6 marks)**

(b) On Figure 2 on page 5 plot a graph of $1/u$ against $1/v$, using 2 cm to represent 1×10^{-2} cm^{-1} on each axis. **(7 marks)**

GO ON TO THE NEXT PAGE

Scanned with CamScanner

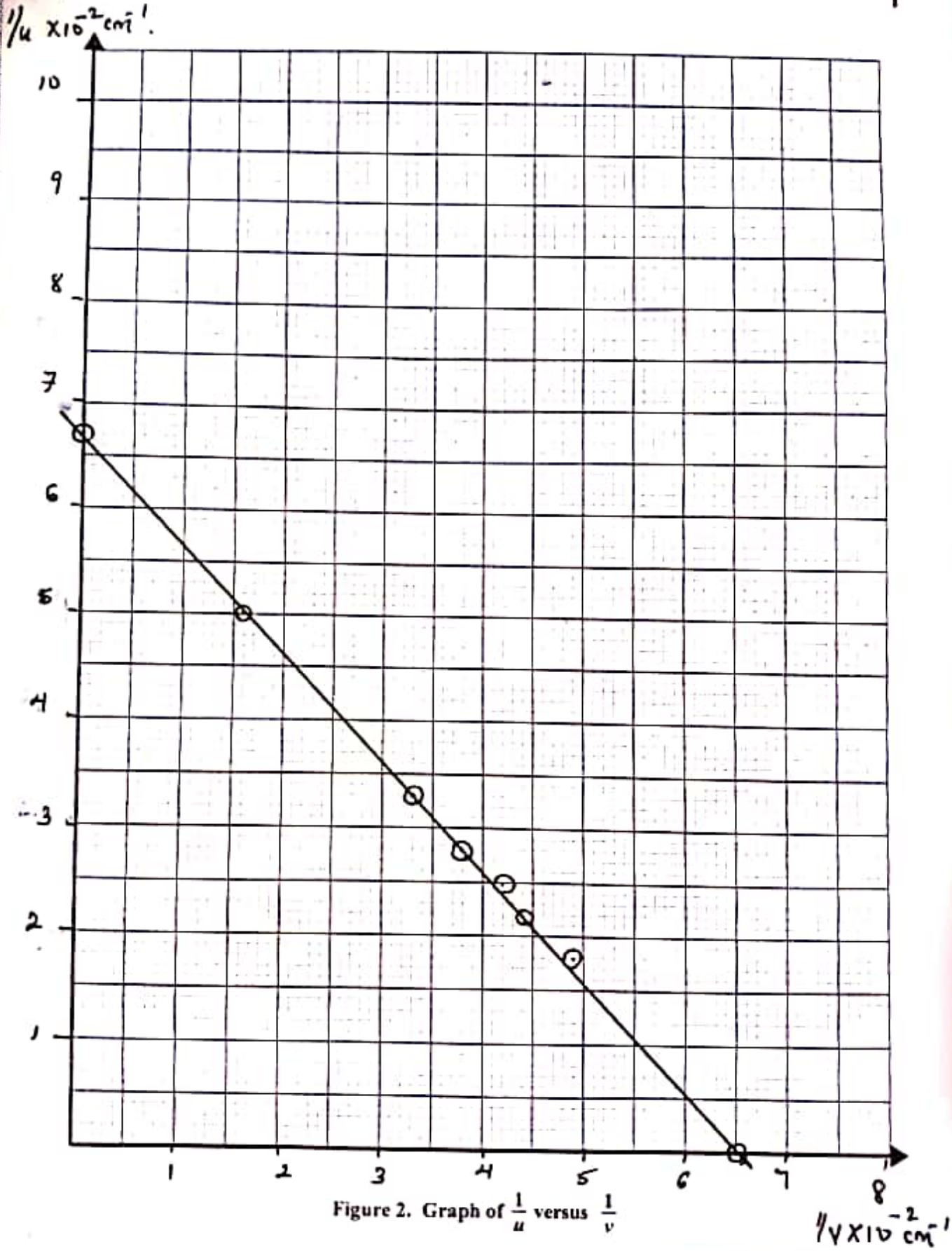


Figure 2. Graph of $\frac{1}{u}$ versus $\frac{1}{v}$

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Scanned with CamScanner

(c) Use the graph to determine the value of v when $1/u$ is $4.0 \times 10^{-2}\,cm^{-1}$.

$$V = \frac{1}{2.6 \times 10^{2}} = \frac{1}{0.026} = 38.5\,cm.$$

(2 marks)

(d) (i) Use the graph to determine the intercepts, c_1 and c_2, on both axes, and calculate the average value, c.

y intercept : 6.7, . x intercept = 6.5 .

Average intercept (c) : $\frac{0.067 + 0.065}{2} = 0.066\,cm.$

(4 marks)

(ii) Use the formula $c = 1/f$ to calculate the focal length, f, of the lens.

$$C = \frac{1}{f} = f = \frac{1}{C} = \frac{1}{0.066} = 15\,cm$$

$$\boxed{f = 15.0\,cm.}$$

(3 marks)

GO ON TO THE NEXT PAGE

Scanned with CamScanner

(e) Complete the following statement:

The principal focus of a concave lens is defined as the ....POINT.... on the principal axis from which rays originally ....parallel and close.... to the principal axis ....from which they appear to diverge.... after passing through the lens. **(3 marks)**

Total 25 marks

Scanned with CamScanner

2. (a) State the 'principle of moments'.

The Sum of the clockwise moments about any point is equal to the sum of the anticlockwise moments about that same point (ROTATIONAL EQUILIBRIUM).

(3 marks)

(b) State another condition that is necessary for a body to be in equilibrium.

The sum of the forces in any direction is equal to the sum of the forces in the opposite direction (TRANSLATIONAL EQUILIBRIUM).

(1 mark)

(c) State the formula for the moment of a force, providing the meaning of each variable and its SI unit.

MOMENT = FORCE X PERPENDICULAR DISTANCE.

$T = F \times D$. (Newton Metre - Nm)

F = FORCE (N).

D = DISTANCE: (m).

(3 marks)

Scanned with CamScanner

(d) Figure 3 shows a girl standing on a uniform plank, AB, of length 3 m and weight 270 N. The plank rests on two scales that are located at P and Q, a distance of 2 m apart. The scales are equidistant from the ends of the plank as shown in Figure 3. The scale at P reads 354 N and the scale at Q reads 417 N.

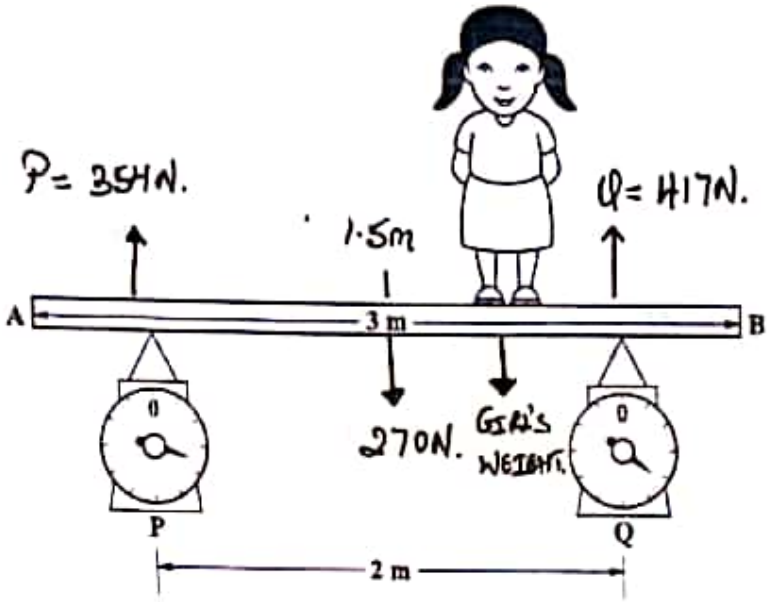


Figure 3. A girl standing on a uniform plank

(i) On Figure 3, draw and label ALL the forces that are acting on the plank.

(2 marks)

(ii) Calculate the girl's weight.

Using the law of Parallel Forces: (Translational Equilibrium)

Sum of Upward forces = Sum. of Downward forces

P + Q = G.O.G. + GIRL'S WEIGHT.

354N + 417N = 270N + GIRL'S WEIGHT. (X_G).

X_G = (771 − 270) N = 501N.

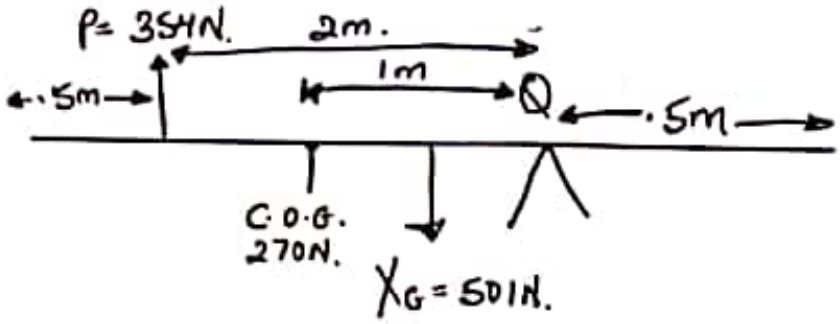
X_G = 501N.

(2 marks)

GO ON TO THE NEXT PAGE

Scanned with CamScanner

(e) By taking moments about Q, calculate the girl's distance from the scale at Q.



Taking Moments About Q.

Clockwise Moments ≡ Counter- Clockwise Moments.

$(354N \times 2m) \equiv (270N \times 1m) + (501 \times x)$.

$708\ Nm \equiv 270 + 501x$

$(708 - 270)\ Nm = 501x$

$438\ Nm = 501x$

$\frac{438\ Nm}{501\ N} = x$

$0.87m. = x$

(4 marks)

∴ Girl's distance from the scale at Q = 0.87m

Total 15 marks

Scanned with CamScanner

DO NOT WRITE IN THIS

3. (a) (i) Define the term 'pressure'.

Pressure can be defined as the force acting perpendicularly per unit area (Nm^{-2}) / Pa

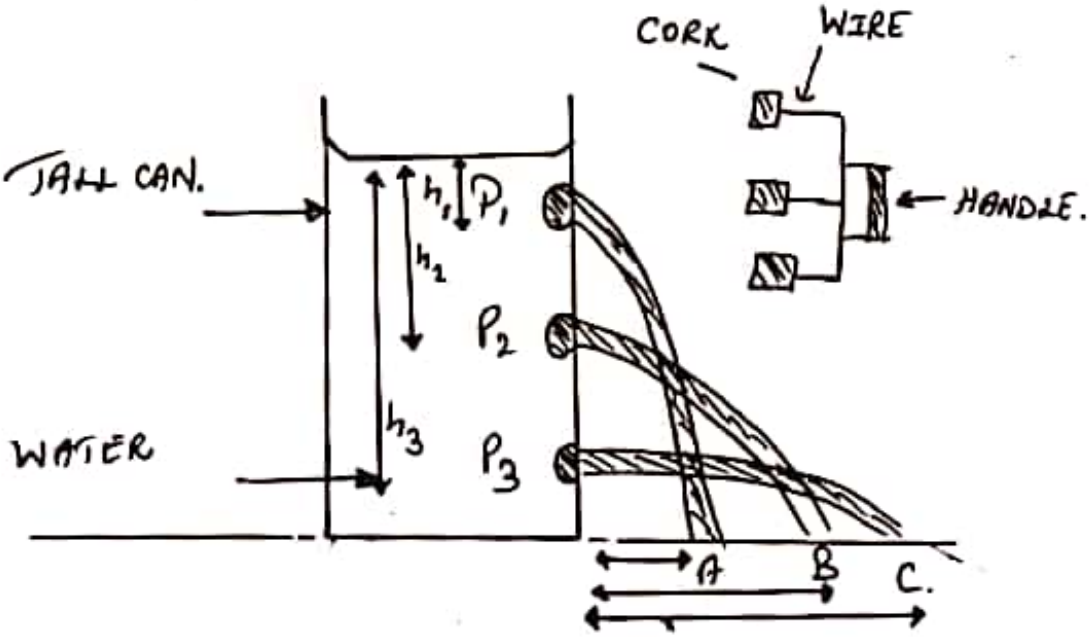
(2 marks)

(ii) State the SI unit of pressure pascal (Pa), in terms of its base units.

Nm^{-2} (Newton per square metre).

(1 mark)

(b) (i) Describe, with the aid of a diagram, how the pressure in a liquid varies with the depth of the liquid.



(i) Use a can with holes at various depths, h

(ii) Fill to the brim with water.

(iii) Pull on the handle so that the corks will be pulled out SIMULTANEOUSLY and observe distances, X. The water at the greater depth squirts out a longer distance therefore has the greatest pressure.

(3 marks)

Scanned with CamScanner

(ii) State the equation for the pressure difference (ΔP) between two points in a liquid, which are a vertical distance, h, apart.

$\Delta P = P_2 - P_1 = \rho g \Delta h.$

(1 mark)

(c) Figure 4 shows a U-tube containing two immiscible liquids, X and Y.

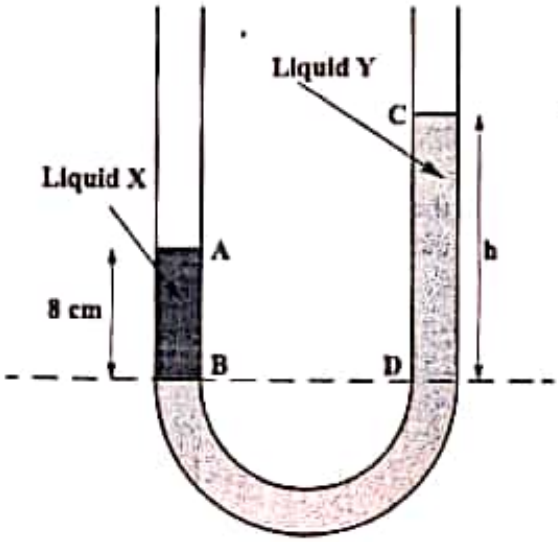


Figure 4. U-tube of immiscible liquids

(i) If the density of Liquid X is 1350 kg m^{-3}, calculate the pressure (in Pa), due to the liquid column AB, which is 8 cm long.

$P = \rho_x g h$

$= 1350 kg m^{-3} \times 10 m s^{-1} \times 0.08 m$

$= 1080 N m^{-2} / Pa$

$\boxed{P_x = 1080 Pa}$

$\rho_x = 1350 kg m^{-3}$

$h = 0.08 m.$

$g = 10 m s^{-1}$

(3 marks)

GO ON TO THE NEXT PAGE

Scanned with CamScanner

(ii) Hence, use your answer to (c) (i) to calculate the height, h, of the liquid column CD, given that the density of liquid Y is 900 kg m^{-3}. **(State any equations used and fully explain the reasoning in obtaining your answer to (c) (ii).)**

[Use g = 10 ms^{-2}]

$\Delta P = \rho g \Delta h.$

$\rho_x > \rho_y$. = The pressure exerted on ρ_y is due to ρ_x.

$P = \rho g \Delta h$

$1080\ Pa = 900\ kgm^{-3} \times 10\ ms^{-2} \times h$

$1080\ Pa = 9000h$

$\frac{1080\ Pa}{9000} = h$

$\boxed{0.12m. = h}$

In, cm: $.12 \times 100 = \underline{12cm.}$

(5 marks)

Total 15 marks

GO ON TO THE NEXT PAGE

Scanned with CamScanner

SECTION B

Answer ALL questions.

4. (a) A young student has just entered secondary school. He was accustomed to having his home-cooked meal warmed up at lunch time in the microwave oven at his primary school. He misses this and suggests to his parents that they buy him a vacuum flask similar to the one shown in Figure 5.

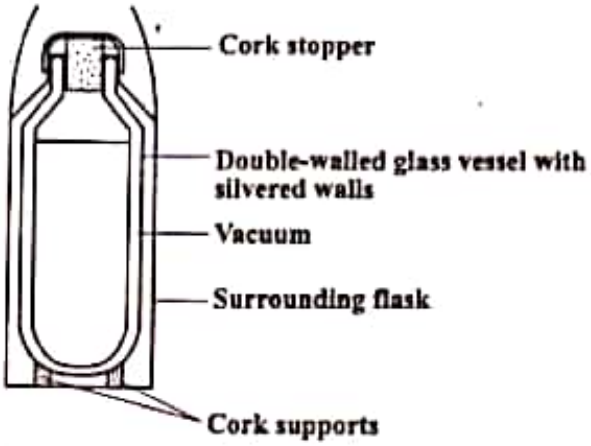


Figure 5. A vacuum flask

Using the information in Figure 5, explain in terms of the principles of thermal energy transfer how EACH of the following sections of the vacuum flask helps to keep the home-cooked meal warm.

(i) The cork stopper

Prevents heat loss by convection and radiation. as it has a low thermal capacity / conductivity.

(ii) The vacuum

Prevents transfer of thermal energy by conduction and convection.

Scanned with CamScanner

(iii) The double-walled glass vessel with silvered walls glass is a poor conductor.

(i) Thermal energy transfer by conduction is reduced

(ii) silvered walls prevents radiation by continuous reflection of heat which keeps the contents warm (6 marks)

(b) In an experiment to determine the specific latent heat of fusion of ice, shown in Figure 6, the following results were obtained.

The initial mass of the water is 120 g. The initial temperature of the water is 35 °C and the initial temperature of the ice is 0 °C.

The final mass of the water is 140 g and the final temperature of the water plus the melted ice is 18 °C.

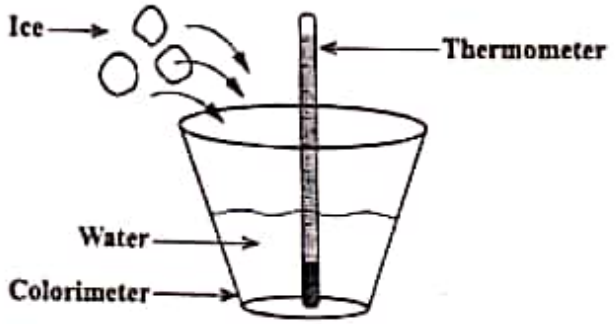


Figure 6. Experiment to determine the specific latent heat of fusion of ice

Assume the container has negligible heat capacity and that there is no net heat transfer with the surroundings during the experiment.

(i) Determine the mass of the melted ice.

140g - 120g = 20g of ice

(1 mark)

(ii) Calculate the heat lost by the water.

$E_H = m_w C_w \Delta\theta$

$= 120g \times 4.2 J g^{-1} K^{-1} \times (35°c - 18°c)$

$120g \times 4.2 J g^{-1} K^{-1} \times 17°c$

$E_H = 8568 J.$

(3 marks)

Scanned with CamScanner

(iii) Write an expression for the TOTAL heat gained by the ice in melting and the melted ice in warming to 18 °C.

$m_i l_f + m_i c\Delta\theta$ where m_i = mass of ice

(2 marks)

(iv) Hence calculate the specific latent heat of fusion of ice.

[Specific heat capacity of water = 4.2 J g^{-1} K^{-1}]

$m l_f + m_i c\Delta\theta = 8568\ J.$

$20g\ l_f + 20g \times 4.2 J g^{-1} C^{-1} \times 18°C = 8568\ J.$

$20 l_f + 1512 J = 8568 J.$

$20 l_f = (8568 - 1512) J.$

$l_f = \frac{7056}{20} = 352.8 \sim 353$

$\boxed{l_f = 353 J g^{-1}}$

(3 marks)

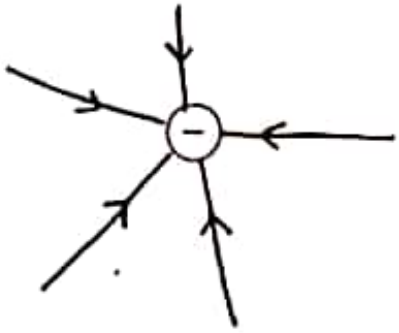
Total 15 marks

GO ON TO THE NEXT PAGE

Scanned with CamScanner

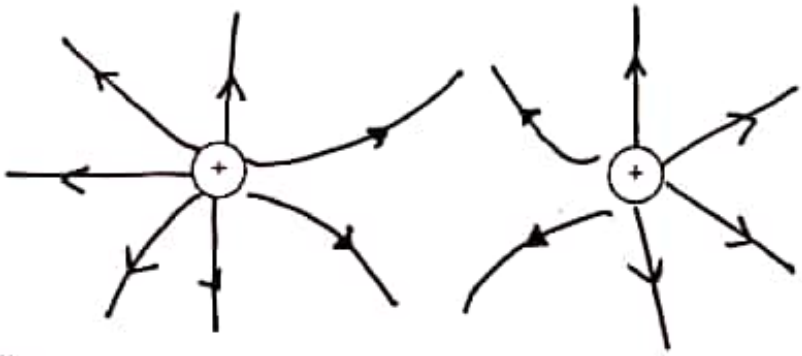
5. (a) Draw the electric fields for EACH of the following arrangements.

(i)



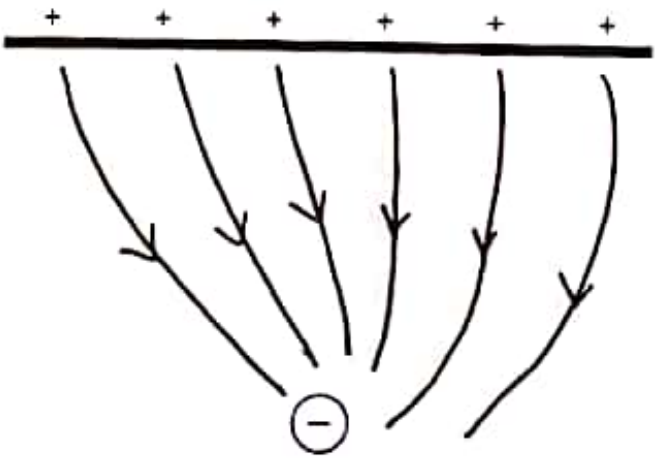
(2 marks)

(ii)



(2 marks)

(iii)



(2 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Scanned with CamScanner

(b) If the number of electrons transferred from a charged rod to earth through a wire is 4×10^8 and the total charge transferred is 0.64 C, calculate

(i) the magnitude of the charge on one electron

$Q = nq$

$q = \frac{Q}{n} = \frac{0.64}{4.0 \times 10^8} = 1.0 \times 10^{-9}$ C.

Charge of 1 electron: 1.0×10^{-9} C.

(3 marks)

(ii) the current that flows through the wire if it takes 0.40 s for all the electrons to flow to earth.

$Q = i \times t$

$i = \frac{Q}{t} = \frac{0.64 C}{.4 s} = 1.6 A$.

(3 marks)

(iii) State the direction in which the charges will flow, if the rod is positively charged.

It will flow towards the positively charged object causing attraction.

(1 mark)

(c) Explain how the static 'build up' of charges on a vehicle could be hazardous.

Fuel running through the pipes can provide the friction needed to create a static charge. If the charge is large, a spark to the earth can happen unexpectedly, potentially igniting a fuse especially when re-fueling vehicles.

(2 marks)

Total 15 marks

GO ON TO THE NEXT PAGE

Scanned with CamScanner

6. (a) Figure 7 shows the experimental arrangement of the apparatus used to carry out the Rutherford's alpha (α) particle scattering experiment to identify the structure of the atom.

A beam (collimated beam) of alpha particles from a radioactive source was fired at a thin piece of gold foil and the angle of deviation of the alpha particles observed.

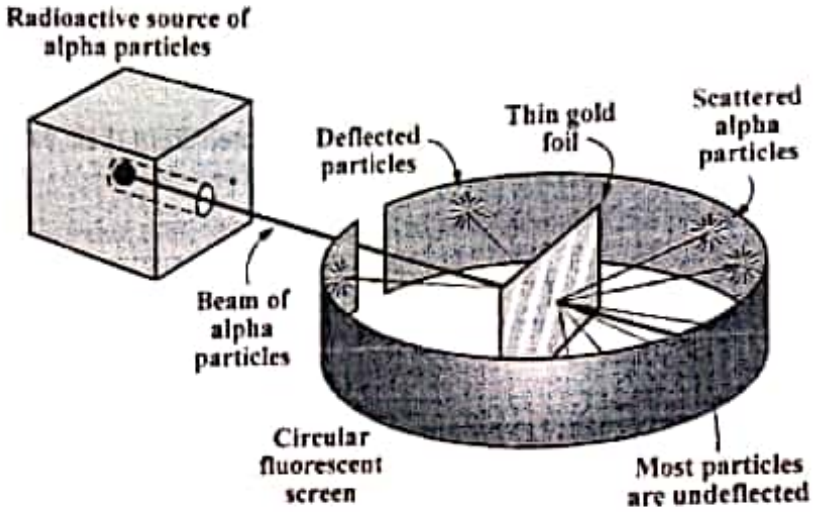


Figure 7. Rutherford's alpha particle scattering experiment

(i) Write the names of the two scientists who assisted Rutherford in performing this experiment.

Mr. HANS GEIGER

Mr. ERNEST MARSDEN.

(2 marks)

(ii) Explain why the experiment was carried out in an evacuated chamber.

The chamber is evacuated because alpha particles are stopped by a few centimetres of air. Any presence of air particles will result in the alpha particles not hitting the gold foil which can adversely affect the desired results.

(2 marks)

Scanned with CamScanner

(b) State the conclusions that were deduced from EACH of the following observations made from the alpha particle scattering experiment.

(i) Most of the alpha particles passed through the gold foil undeviated.

Most of the gold atoms is empty space.

(1 mark)

(ii) A few alpha particles were deflected from their path but continued through the gold foil.

The presence of a positively charged entity (nucleus) 99.9% of mass is concentrated in a small region of the atom's centre.

(1 mark)

(iii) A smaller number of alpha particles rebounded.

The nucleus is extremely dense but highly charged as exemplified by the rebounded alpha-particles.* ELECTRONS ARE FOUND IN THE SPACE OUTSIDE THE NUCLEUS.*

(1 mark)

(c) Describe the Rutherford model of the atom, based on the conclusions of the experiment stated in (b).

The Rutherford model described the atom as a tiny dense, positively charged core called the nucleus, in which nearly all the mass is concentrated around which light negative constituents called electrons circulate at some distance, much like planets revolving around the sun.

(2 marks)

Scanned with CamScanner

(d) Energy is produced within stars by the process of fusion where alpha particles help convert beryllium into carbon, according to the equation below.

$$^{4}_{2}He + ^{9}_{4}Be \rightarrow ^{1}_{0}n + ^{12}_{6}C$$

(i) In the equation above, which symbol is used to represent the alpha particle?

$^{4}_{2}He = \alpha$ (Alpha Particle).

(1 mark)

(ii) The following table shows the mass of each nuclei in the process above.

$^{4}He_{2}$	4.00260 u
$^{9}Be_{4}$	9.01212 u
$^{1}n_{0}$	1.00867 u
$^{12}C_{6}$	12.0000 u

Calculate the loss in mass, Δm, in kg, due to the reaction.

[1u = 1.66 × 10^{-27} kg]

Loss in Mass (Mass Defect)

$$^{4}_{2}He + ^{9}_{4}Be \longrightarrow ^{1}_{0}n + ^{12}_{6}C$$

$$(4.00260 + 9.0121)u \longrightarrow (1.00867 + 12.0000)u$$

$$\Delta m_u = 13.01472 \rightarrow 13.00867$$

$$\Delta m_u = 13.00867 - 13.01472 = 6.05 \times 10^{-3} u.$$

$$1u = 1.66 \times 10^{-27} kg.$$

$$\therefore 6.05 \times 10^{-3} u = 6.05 \times 10^{-3} \times 1.66 \times 10^{-27} kg$$

$$\boxed{\Delta m. = 1.0043 \times 10^{-29} kg.}$$

(2 marks)

Scanned with CamScanner

(iii) Calculate the energy equivalence of the mass in (d) (ii).

[$c = 3.00 \times 10^8\ ms^{-1}$]

$$\Delta E = \Delta m c^2$$

$$\Delta E = 1.00743 \times 10^{-29}\ kg \times 3.0 \times 10^8\ ms^{-1} \times 3.0 \times 10^8\ ms^{-1}$$

$$\boxed{\Delta E = 9.0387 \times 10^{-13}\ J.}$$

(3 marks)

Total 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

Scanned with CamScanner